

GrowBikeNet: A Python package to grow urban bicycle networks

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Summary

GrowBikeNet is a Python package that grows an urban bicycle network from scratch or from an existing bicycle network, as a decision support tool for urban planners or to create a compelling vision for urban cycling. A bicycle network is a connected set of protected bicycle infrastructure elements that span a large part of a city, providing safe and direct travel for cyclists. The software downloads and pre-processes street and bicycle network data and points of interests from OpenStreetMap via OSMnx or from custom local data, prepares seed points to which the network should connect, grows a network and routes it on the streets, orders the edges, exports results and plots. The software is available at <https://github.com/BikeNetKit/GrowBikeNet>, functionality is interactively explorable at <https://bikenetkit.org/growbikenet>.

Statement of need

Networks of well-connected, protected bicycle tracks are recognized as fundamental urban infrastructure that spurs cycling. More cycling and less cars makes cities more livable and sustainable, providing massive public health and environmental benefits (Brand et al., 2021; Gössling et al., 2019; Miner et al., 2024; Reynolds et al., 2009). However, the planning and implementation of a bicycle network is a tedious political and technical process, requiring extensive stakeholder involvement and careful consideration of various factors (de Groot, 2016). New computational tools and data sets, such as the OSMnx Python library (Boeing, 2017) and OpenStreetMap (Haklay & Weber, 2008), respectively, provide the opportunity to boost this process by allowing to easily process and visualize urban street networks computationally, and to run simulations of growing subnetworks on top of them. These opportunities have led to several recent computational studies on bicycle networks (Olmos et al., 2020; Szell et al., 2022; Vybornova et al., 2023). Yet, despite ample individual efforts and a beginning consolidation of the field (Diogo Pinto et al., 2026), none of these research projects have led to an open-source, general-use software package for simulated bicycle network growth and planning, visualization, and analysis. GrowBikeNet fills this gap.

The software provides benefits for at least two use cases. First, in urban planning, it provides urban planners and cycling advocates with a quick way to generate city-spanning bicycle network blueprints for their cities and general statistics relevant to the planning process. As GrowBikeNet allows local data imports and has ample settings to configure, it can simulate various scenarios adapted to local contexts. These blueprints can serve as a vision for potential future city development or they can be further refined. After exporting the generated networks to a geographic information system (GIS), detailed modifications could be made, for example with open tools like QGIS (Graser et al., 2025) or A/B Streets which also allows traffic simulations (Carlino et al., 2024). This quick feedback reduces the time and resources required

to manually plan bicycle networks which in turn can accelerate sustainable urban development. Second, GrowBikeNet enables researchers to conduct large-scale studies across multiple cities or regions, providing valuable insights into the potential impacts of bicycle networks at a broader scale. In both cases, GrowBikeNet can help to identify best practices, algorithmic approaches, and strategies for bicycle network implementation.

State of the field

The field of computational bicycle network design is still small, but has grown in the last few years with several new papers per year, and some dozen papers in total, warranting a literature review (Diogo Pinto et al., 2026). Summarizing Diogo Pinto et al. (2026), the field can be categorized in different ways. On one hand is the methodological distinction between graph-based and GIS-based approaches. On the other hand are the different field-specific approaches ranging from network science to transport planning, which often correspond to the first categorization, but not always. GrowBikeNet falls into the graph-based, network science approach, but it aims to bridge the gap from originally exclusive network-structural considerations (Szell et al., 2022) to real applicability for city planning.

When it comes to the openness of data and algorithms, unfortunately many existing projects are closed source and thus not reproducible nor extendable by the community. Notable exceptions are, for example, Lovelace et al. (2017), or our own efforts like Lonardi et al. (2025), Vybornova et al. (2025), or Sebastiao et al. (2026). Another problem is the one-off nature of raw, unmaintained research algorithms which makes their further development difficult even when they are open-source, including the original predecessor algorithm to GrowBikeNet (Szell et al., 2022). To this suboptimal state of the field GrowBikeNet aspires to be a counterpoint: Packaged via pypi and conda, and using the AGPL license which forces openness and reproducibility.

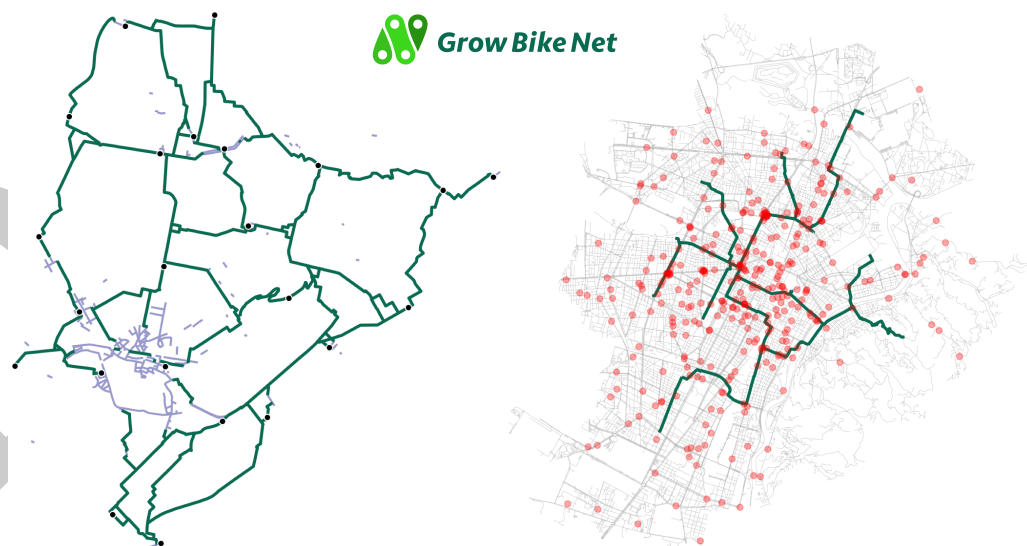


Figure 1: Example results of GrowBikeNet. Left: Growing a full bicycle network (green) for Athens, Greece, with default settings, incorporating existing bicycle infrastructure (purple). Seed points (black) are snapped to both substantial elements of the existing bicycle network and to a grid. The interactive version of this map is available at https://bikenetkit.org/growbikenet/athens_gr. Right: The first 40km of a growing bicycle network (green) for Turin, Italy, reprioritized by traffic crash data (red) to first develop through crash hotspots. The existing bicycle network is ignored and seed points are not shown for illustration purposes, the street network is shown in grey. Map data from OpenStreetMap. Top: BikeNetKit logo with GrowBikeNet wordmark.

66 Software design

67 Operationalizing bicycle network planning concepts

68 The concept of a bicycle network has no agreed upon definition (Schön et al., 2024).
69 GrowBikeNet defines it as the set of infrastructure elements (implemented by physical and/or
70 legal means) which allows people of all ages and demographics to safely cycle - both subjectively
71 and objectively. This definition is broad and somewhat vague on purpose, as the microscopic
72 implementation details (raised curbs, bollards, bicycle streets, etc.) should be up to the local
73 context and informed by local domain knowledge. Thus, GrowBikeNet concerns itself first and
74 foremost with the topological foundation of a bicycle network, which is its network structure.

75 GrowBikeNet's bicycle network design philosophy was developed by Szell et al. (2022) and
76 is based on the idea of linking seed points, inspired by the Dutch CROW Design manual for
77 bicycle traffic (de Groot, 2016). These are points in a city that shall be connected by protected
78 bicycle infrastructure, yielding edges in a bicycle network, see Fig. 1 (left) for an example. The
79 choice and placement of seed points is crucial. To build a well-covering bicycle network, seed
80 points should cover most of the city. By default, in GrowBikeNet seed points are arbitrary,
81 city-spanning points on a grid, snapped to the street network, but GrowBikeNet also allows
82 to assign actual preset points of interests such as railway stations or schools, or arbitrary
83 OpenStreetMap tags or custom points of interest, going beyond Szell et al. (2022). Another
84 important extension to Szell et al. (2022) is the option to not start growing from scratch but
85 to account for the existing bicycle network following Folco et al. (2023), by placing seed points
86 on the existing bicycle network first.

87 Once seed points are set and snapped to the street network, they need to be linked. By default,
88 GrowBikeNet triangulates or quadrangulates the seed points automatically depending on street
89 network structure (Boeing, 2019), which is another extension of Szell et al. (2022), building a
90 triangle or square-shaped link structure between the seed points. Once the linking process is
91 finished, an abstract, unrouted network between the seed points is established. GrowBikeNet
92 then builds a bicycle network on this network's edges, routed on the street network. All network
93 handling is done by NetworkX (Hagberg et al., 2008).

94 After routing, GrowBikeNet orders the edges in this potential bicycle network. The ordering
95 is important, as it informs a city which edges to implement first to arrive at a functional
96 network early. The default ordering is via betweenness centrality which is a network centrality
97 measure approximating flow. By ordering edges like this, the first built edge is the one with
98 highest expected flow of cyclists. The current package version provides the additional ordering
99 options by closeness centrality or randomly. As was shown by Szell et al. (2022), ordering by
100 betweenness centrality is superior to random ordering, and usually also more adequate than
101 ordering by closeness centrality.

102 Finally, edges are rerouted with weights to account for edges proposed in earlier steps, and
103 edge overlaps between different steps are removed. These two postprocessing steps get rid
104 of many artefacts observed from the raw algorithm of Szell et al. (2022) and lead to a more
105 realistic network structure.

106 Custom data for re-prioritizing growth and city definition

107 Additionally to its default mode of operation, GrowBikeNet implements the weighted ordering
108 of Folco et al. (2023), which enables to re-prioritize the growth order of edges through an
109 imported point and/or trip data set, making the simulation more data-driven and realistic.
110 Example data sets include traffic crashes or empirical mobility flows, demonstrated for the city
111 of Turin, Italy in Fig. 1 (right).

112 While GrowBikeNet builds heavily on OpenStreetMap and uses OpenStreetMap data in its
113 default mode of operation, its data loading options allow operating with any data source as

114 long as it follows the same data structure as OpenStreetMap data. This flexibility allows to
115 define a street or existing bicycle network with network data other than from OpenStreetMap,
116 or to limit or extend the specific street types on which the bicycle network is allowed to grow.

117 **Software implementation**

118 GrowBikeNet's software design is based on three core principles: (1) to provide a user-friendly
119 experience of executing one function with ample progress feedback and usage documentation
120 ([GrowBikeNet Contributors, 2026](#)), (2) to use community tools and standards, e.g.,
121 GeoPandas/shapely ([Fleischmann et al., 2026](#)) for handling spatial aspects, OSMnx/NetworkX
122 for handling networks ([Boeing, 2017](#); [Hagberg et al., 2008](#)), and (3) high customizability
123 by making most design decisions configurable via settings and constants and by supporting
124 custom data imports (seed points, street and bicycle networks, city boundary, trip or point
125 data). Data can be imported and exported in common geospatial formats such as shape files
126 (.shp), GeoJSON (.geojson), or GeoPackage (.gpkg).

127 Performance of network algorithms, for example NetworkX's Python implementation of shortest
128 paths, is generally not an issue, as the number of shortest paths to calculate is limited. One
129 performance bottleneck is data download via OSMnx and its underlying Overpass API for
130 the initial data download. This bottleneck is mitigated by OSMnx using JSON caching for
131 subsequent data uses and by GrowBikeNet's option to import local data sets. The step
132 of removing overlapping edges at the end of GrowBikeNet's procedure is computationally
133 complex, therefore it utilizes shapely's highly optimized GEOS library written in C++. For a
134 medium-sized city and an off-the-shelf laptop, one GrowBikeNet run takes below one minute,
135 for large cities not more than a few minutes.

136 **Embedding in BikeNetKit**

137 GrowBikeNet is part of the BikeNetKit family of bicycle network toolkits and utilities.
138 GrowBikeNet's typical use case is underdeveloped cities (in terms of bicycle network
139 development), making it well usable for most cities on the planet. Other BikeNetKit tools,
140 like superblockify ([Büth et al., 2024](#)), have complementary use cases. Shared functions
141 between different BikeNetKit tools are planned to be consolidated and imported from the
142 utility package BikeNetLib in the future.

143 **Visualization**

144 GrowBikeNet allows to plot each step in the simulated network growth using matplotlib.
145 GrowBikeNet also provides a script to use all plots as frames for generating a video via openCV.
146 Visual aspects like colors, line widths, etc., are configurable. Example results are shown in
147 Fig. 1. Precomputed results of GrowBikeNet for over 400 European cities can be interactively
148 explored at <https://bikenetkit.org/growbikenet>.

149 **License and philosophy**

150 The AGPL license serves two purposes. First, it ensures the essential scientific standard of
151 reproducibility of GrowBikeNet and of all its derived versions. Second, it positions GrowBikeNet
152 as a public good rather than a proprietary consultancy tool for extracting city budgets.
153 GrowBikeNet provides free, open-source software in a community effort that can be extended
154 at will, and which cities and proactive citizens can use directly to explore and revisit different
155 scenarios dynamically and in a data-driven way.

Research impact statement

The software's underlying method has already been applied in several research projects. Its original version was developed by Szell et al. (2022) who studied 62 cities to identify optimal growth strategies for urban bicycle networks. Follow-up projects include, for example, Folco et al. (2023) who extended the approach to point and trip data, Basilone et al. (2025) who applied the approach to road widths, Sebastiao et al. (2026) who followed up on a systematic investigation of link ordering, ongoing research by Larkin et al. (2026) who incorporate low-traffic neighborhoods, or Master theses like Pessoa (2026). With increased focus on reducing car-dependence and the negative impacts of urbanization on climate change (Brand et al., 2021; Mattioli et al., 2020; Satterthwaite, 2009), the need for sustainable urban planning tools like GrowBikeNet will only increase.

AI usage disclosure

No generative AI tools were used in the development of this software, the writing of this manuscript, or the preparation of supporting materials.

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References

- Basilone, R., Bruno, M., Piaget Monteiro Melo, H., Avalor, M., & Loreto, V. (2025). Road-Width-Aware network optimisation for bike lane planning. *Journal of Physics: Complexity*, 6(3), 035008. <https://doi.org/10.1088/2632-072X/adf683>
- Boeing, G. (2017). OSMnx: New methods for acquiring, constructing, analyzing, and visualizing complex street networks. *Computers, Environment and Urban Systems*, 65, 126–139. <https://doi.org/10.1016/j.compenvurbsys.2017.05.004>
- Boeing, G. (2019). Urban spatial order: Street network orientation, configuration, and entropy. *Applied Network Science*, 4(1), 67. <https://doi.org/10.1007/s41109-019-0189-1>
- Brand, C., Dons, E., Anaya-Boig, E., Avila-Palencia, I., Clark, A., De Nazelle, A., Gascon, M., Gaupp-Berghausen, M., Gerike, R., Götschi, T., Iacorossi, F., Kahlmeier, S., Laeremans, M., Nieuwenhuijsen, M. J., Pablo Orjuela, J., Racioppi, F., Raser, E., Rojas-Rueda, D., Standaert, A., ... Int Panis, L. (2021). The climate change mitigation effects of daily active travel in cities. *Transportation Research Part D: Transport and Environment*, 93, 102764. <https://doi.org/10.1016/j.trd.2021.102764>
- Büth, C. M., Vybornova, A., & Szell, M. (2024). Superblockify: A Python Package for Automated Generation, Visualization, and Analysis of Potential Superblocks in Cities. *Journal of Open Source Software*, 9(100), 6798. <https://doi.org/10.21105/joss.06798>
- Carlino, D., Kirk, M., Smith, A., dcarlino, Konieczny, M., Kott, G., Bruce, Nissar, J., Nederlof, T., Steinberg, V., Lovelace, R., Ilias, Nebeker, J., Sam, Orestis, Dejean, M., Shenfield, M., Schimek, N., Foucault, M., ... Huston, K. (2024). *A-b-street/abstreet: Night markets at -15C* (Version v0.3.49). Zenodo. <https://doi.org/10.5281/zenodo.10476253>
- de Groot, R. (2016). *Design Manual for Bicycle Traffic*. CROW. ISBN: 978-90-6628-659-7

- 199 Diogo Pinto, J., Pinheiro, F. L., & Bação, F. (2026). Frameworks for prioritizing cycling network
200 investments: A systematic literature review. *Transportation Research Interdisciplinary*
201 *Perspectives*, 35, 101782. <https://doi.org/10.1016/j.trip.2025.101782>
- 202 Fleischmann, M., Van Den Bossche, J., Jordahl, K., Richards, M. J., McBride, J., Wasserman,
203 J., Ward, B., & Wolf, L. J. (2026). GeoPandas: Fundamental data structures for vector
204 spatial data in Python. *Computers, Environment and Urban Systems*, 130, 102495.
205 <https://doi.org/10.1016/j.compenvurbsys.2026.102495>
- 206 Folco, P., Gauvin, L., Tizzoni, M., & Szell, M. (2023). Data-driven micromobility network
207 planning for demand and safety. *Environment and Planning B: Urban Analytics and City*
208 *Science*, 50(8), 2087–2102. <https://doi.org/10.1177/23998083221135611>
- 209 Gössling, S., Choi, A., Dekker, K., & Metzler, D. (2019). The Social Cost of Automobility,
210 Cycling and Walking in the European Union. *Ecological Economics*, 158, 65–74. <https://doi.org/10.1016/j.ecolecon.2018.12.016>
- 212 Graser, A., Sutton, T., & Bernasocchi, M. (2025). The QGIS project: Spatial without
213 compromise. *Patterns*, 6(7), 101265. <https://doi.org/10.1016/j.patter.2025.101265>
- 214 GrowBikeNet Contributors. (2026). *GrowBikeNet - Usage*. [https://docs.bikenetkit.org/](https://docs.bikenetkit.org/GrowBikeNet/usage/)
215 [GrowBikeNet/usage/](https://docs.bikenetkit.org/GrowBikeNet/usage/)
- 216 Hagberg, A. A., Schult, D. A., & Swart, P. J. (2008). Exploring Network Structure,
217 Dynamics, and Function using NetworkX. In G. Varoquaux, T. Vaught, & J. Millman
218 (Eds.), *Proceedings of the 7th Python in Science Conference* (pp. 11–15). <https://doi.org/10.25080/tcww9851>
- 220 Haklay, M., & Weber, P. (2008). OpenStreetMap: User-Generated Street Maps. *IEEE*
221 *Pervasive Computing*, 7(4), 12–18. <https://doi.org/10.1109/MPRV.2008.80>
- 222 Larkin, C., Vybornova, A., Sebastiao, C., Robson, C., & Szell, M. (2026). Implementing low
223 traffic neighbourhoods to the expansion of cycle networks. *The 34th Annual GIS Research*
224 *UK (GISRUK) Conference*. <https://doi.org/10.5281/zenodo.19390771>
- 225 Lonardi, A., Szell, M., & De Bacco, C. (2025). Cohesive urban bicycle infrastructure design
226 through optimal transport routing in multilayer networks. *Journal of The Royal Society*
227 *Interface*, 22(223), 20240532. <https://doi.org/10.1098/rsif.2024.0532>
- 228 Lovelace, R., Goodman, A., Aldred, R., Berkoff, N., Abbas, A., & Woodcock, J. (2017). The
229 Propensity to Cycle Tool: An open source online system for sustainable transport planning.
230 *Journal of Transport and Land Use*, 10(1). <https://doi.org/10.5198/jtlu.2016.862>
- 231 Mattioli, G., Roberts, C., Steinberger, J. K., & Brown, A. (2020). The political economy of
232 car dependence: A systems of provision approach. *Energy Research & Social Science*, 66,
233 101486. <https://doi.org/10.1016/j.erss.2020.101486>
- 234 Miner, P., Smith, B. M., Jani, A., McNeill, G., & Gathorne-Hardy, A. (2024). Car harm: A
235 global review of automobility's harm to people and the environment. *Journal of Transport*
236 *Geography*, 115, 103817. <https://doi.org/10.1016/j.jtrangeo.2024.103817>
- 237 Olmos, L. E., Tadeo, M. S., Vlachogiannis, D., Alhasoun, F., Espinet Alegre, X., Ochoa, C.,
238 Targa, F., & González, M. C. (2020). A data science framework for planning the growth
239 of bicycle infrastructures. *Transportation Research Part C: Emerging Technologies*, 115,
240 102640. <https://doi.org/10.1016/j.trc.2020.102640>
- 241 Pessoa, M. (2026). *Assessing algorithmic versus municipal planning of Lisbon's bicycle network*
242 [Master's thesis]. IT University of Copenhagen.
- 243 Reynolds, C. C., Harris, M. A., Teschke, K., Cipton, P. A., & Winters, M. (2009). The impact
244 of transportation infrastructure on bicycling injuries and crashes: A review of the literature.
245 *Environmental Health*, 8(1), 47. <https://doi.org/10.1186/1476-069X-8-47>

- 246 Satterthwaite, D. (2009). The implications of population growth and urbanization for
247 climate change. *Environment and Urbanization*, 21(2), 545–567. <https://doi.org/10.1177/0956247809344361>
248
- 249 Schön, P., Heinen, E., & Manum, B. (2024). A scoping review on cycling network connectivity
250 and its effects on cycling. *Transport Reviews*, 44(4), 912–936. <https://doi.org/10.1080/01441647.2024.2337880>
251
- 252 Sebastiao, C., Vybornova, A., Vierø, A. R., Aiello, L. M., & Szell, M. (2026). The trade-off
253 between directness and coverage in transport network growth. *Applied Network Science*,
254 11(1), 34. <https://doi.org/10.1007/s41109-026-00792-5>
- 255 Szell, M., Mimar, S., Perlman, T., Ghoshal, G., & Sinatra, R. (2022). Growing urban bicycle
256 networks. *Scientific Reports*, 12(1), 6765. <https://doi.org/10.1038/s41598-022-10783-y>
- 257 Vybornova, A., Cunha, T., Gühnemann, A., & Szell, M. (2023). Automated Detection
258 of Missing Links in Bicycle Networks. *Geographical Analysis*, 55(2), 239–267. <https://doi.org/10.1111/gean.12324>
259
- 260 Vybornova, A., Vierø, A. R., Krogh Hansen, K., & Szell, M. (2025). BikeNodePlanner: A
261 data-driven decision support tool for bicycle node network planning. *Environment and*
262 *Planning B: Urban Analytics and City Science*, 52(7), 1771–1780. <https://doi.org/10.1177/23998083251355999>
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